

- 1 The Earth may be assumed to be a uniform sphere of radius 6400 km and mass 6.02×10^{24} kg.
- (a) A 50.0 kg boy is standing still on a flat ground located at latitude 35.6° north of the Equator, somewhere in Japan, as shown in Fig 1.1.

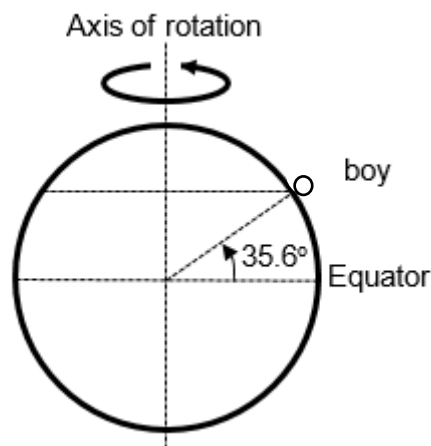


Fig. 1.1

- (i) Draw and label all the forces acting on the boy on Fig. 1.2.

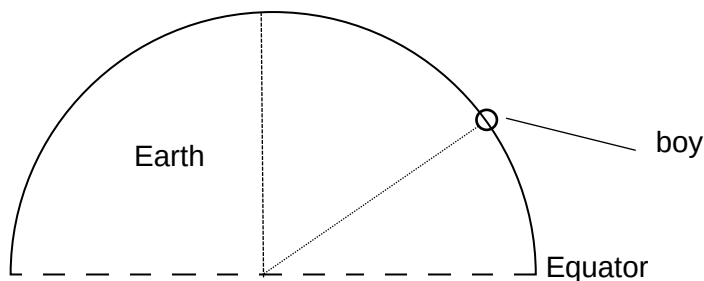


Fig. 1.2

- (ii) The boy now puts a weighing scale under him to read his weight from the scale.

Explain whether there is a difference in the readings on the weighing scale if the boy stands on the same weighing scale at the North pole.

.....

.....

.....

.....

[2]

[Turn over]

- (b) A satellite orbiting the Earth with a period of 24 hours and flew directly above the boy from west to east. The satellite is under the influence of gravitational force alone.
- (i) Determine the height of the satellite above the boy.

height = m [3]

- (ii) Explain whether it is a geostationary satellite above the boy.

.....

.....

..... [1]

DO NOT WRITE IN THIS
MARGIN

DO NOT WRITE IN THIS
MARGIN

[2 (a) The first law of thermodynamics may be given by the expression